

The Legacy of Daniel Kahneman: A Personal View

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Daniel Kahneman and Amos Tversky's joint papers from the 1970s and 1980s have inspired many, including myself. These articles magically turned statistical thinking—previously a niche interest—into a major research focus. Kahneman and Tversky revived the concept of heuristics, which had largely been forgotten at the time, and played a pivotal role in bringing psychology to the attention of economics and other social sciences. I was also deeply influenced by Tversky's seminal work on the foundations of measurement, which inspired my first book on modeling (Gigerenzer 1981).

In their joint work, known as the heuristics-and-biases program, Kahneman and Tversky argued that human judgment systematically deviates from the norms of probability and logic, resulting in predictable cognitive biases. These biases were attributed to heuristics—mental shortcuts—which led to a broader narrative in behavioral economics and psychology that emphasized human fallibility in decision-making.

The heuristics-and-biases program sparked intense debate on the nature of human rationality. This debate placed me in direct opposition to Kahneman and Tversky, with Kahneman referring to me in *Thinking, Fast and Slow* (Kahneman 2011, 449) as “our most persistent critic”. At stake was more than simply an academic disagreement—it was a fundamental clash of worldviews on rationality, intelligence, the standards for sound judgment, and their implications for policy-making.

In this article, I will focus on Kahneman's legacy: the contributions made and the opportunities missed. After discussing these, I reflect on the challenges we faced in our exchanges, both as fellow scientists and fellow human beings. It is of course impossible to discuss Kahneman without referring to Tversky as well, given that much of Kahneman's early work was written with him. At the same time, although Kahneman and Tversky are often spoken of as a unit, they were markedly different, both personally and scientifically. This contrast was part of their success. For more than a dozen years, they formed a highly influential partnership. However, their differing personalities also led to the eventual end of their

collaboration. In *The Undoing Project*, biographer Michael Lewis describes how their close friendship began to unravel in the late 1980s, with Kahneman later remarking: “I sort of divorced him” (Lewis 2017, 338). Their intellectual partnership briefly reignited in the mid-1990s, when Tversky convinced Kahneman to co-author a reply to my critique (Kahneman and Tversky 1996, Gigerenzer 1996). Sadly, Tversky passed away that same year.

BEFORE KAHNEMAN AND TVERSKY

Humankind was slow to discover mathematical probability. The delay was so striking that Hacking (1975) called it a “scandal of philosophy”, dating its emergence to 1654, when Blaise Pascal and Pierre Fermat exchanged letters on gambling problems. Psychologists were equally slow. Since the 19th century, significant figures such as Wilhelm Wundt had debated the relationship between intuition and logic, but overlooked probability. It was not until the 1950s and 60s that psychologists began studying statistical intuitions.

The first systematic investigation was conducted by Swiss psychologists Jean Piaget and Bärbel Inhelder in *The Origin of the Idea of Chance in Children* ([1951]1975). They presented children of various ages with random devices and concluded that by age 12, children’s intuitions begin to align with the laws of probability. For instance, children understand the law of large numbers and, by age eight, the conjunction rule (Inhelder and Piaget 1964). Around the same time, Egon Brunswik (1955) at the University of California, Berkeley, envisioned the perceptual system as an intuitive statistician, one that calculates correlations and multiple regressions.

At the University of Michigan, Ward Edwards (1962, 1968), often referred to as the founder of behavioral decision theory, compared people’s judgments with Bayes’ rule and found that people are generally good Bayesians, albeit conservative ones. Similarly, in a review of 110 experimental articles, Peterson and Beach (1967, 43) concluded that the laws of statistics provide “a good first approximation for a psychological theory of inference”. Their article was—apart from the gender bias of the time—aptly entitled “*Man as intuitive statistician*”.

By the end of the 1960s, more than 100 experiments had demonstrated that on the whole, perception and judgment followed the laws of probability. That did not mean that errors do not occur. Back in the 18th century, Pierre-Simon Laplace noted the gambler’s fallacy, and later on, Brunswik understood systematic errors to be inevitable, given that the

perceptual system lacks direct access to reality and must rely on imperfect cues to make inferences.

Taken together, this body of research confirmed what Laplace and other Enlightenment thinkers had long believed: Probability and reasoning are two sides of the same coin. In Laplace's words, "The theory of probability is at bottom nothing more than good sense reduced to a calculus" (Laplace [1814] 1951, 196). This intimate relation between reasoning and probability was part of a broader vision, that mathematics is not an abstract entity but one intricately woven into reality (Daston 1988). Just as Euclidean geometry captures the space we live in, probability theory captures human reasoning. *Homo sapiens* is, in many ways, an intuitive statistician.

KAHNEMAN CHANGES METHOD AND MESSAGE

It is often said that in the 1970s, Kahneman and Tversky entered the field and revealed that human judgment is riddled with systematic biases. But that account is not entirely accurate. Tversky was already in the field, and in fact one of those who initially argued that intuition largely conforms to the laws of probability. He had studied probability learning while working on his PhD under the supervision of Edwards (Tversky and Edwards 1966). Kahneman (2002a) attended a talk by his younger colleague Tversky and found the idea that people are conservative Bayesians implausible. As quoted by Lewis (2017, 148), Kahneman remarked: "I knew I was a lousy intuitive statistician," adding: "I really didn't think I was stupider than anyone else". Never one to mince words, he faulted Edwards and his colleagues for what he called "their inability to face the evidence of their own folly," dismissing their conclusion that people are conservative Bayesians as "worse than meaningless" (Lewis 2017, 148–149).

Initially, Tversky appears to have been torn between his mentor and his friend. It is conceivable that, had he not encountered Kahneman, Tversky might have continued developing Edwards' research program. Instead, he and Kahneman joined forces and set out to prove Tversky's mentor wrong.

To refute the intuitive statistician required a change in the way experiments were conducted. Experimental researchers—from Piaget and Inhelder to Edwards—had used random devices such as urns and balls, and given participants plenty of time to acquire experience with these devices. For instance, Tversky and Edwards (1966) studied each of 24 participants individually, each seated in front of a random device, with 1,000 trials to learn and make predictions. A typical session lasted about an hour.

Kahneman introduced a radically different method, a practice he called “the psychology of single questions” (2022a). In this approach, participants were presented with a brief text problem instead of a physical random device, had no opportunity to learn or acquire experience, and were tested in large groups instead of individually (Heukelom 2012). A single question—a brief text problem—could be answered by dozens of undergraduates in a couple of minutes.

Equally important, Kahneman also extended the content of the text problems beyond random devices to now famous problems such as whether Linda is a bank teller, Jack is an engineer, or whether a hit-and-run accident was caused by the Blue Taxi company. Pouring real-world content into the dry urns and balls made the problems more interesting and memorable. But that created a new challenge: Is there still a unique correct answer to each problem, independent of the content? Most practicing statisticians start by investigating the content of a problem, work out a set of assumptions, and then build a statistical model. Kahneman and Tversky started at the opposite end. They chose a convenient logical principle, such as Bayes’ rule or the conjunction rule, and filled in some real-world content but assumed that content is irrelevant to determining the correct answer. If people thought about the content and came up with a different answer, Kahneman and Tversky attributed the discrepancy to a flaw in people’s judgment, never to a flaw in their own norm. This firm faith in content-blind norms eventually became the Achilles’ heel of the program.

These changes in experimental protocol—no opportunity for learning, single questions, quick studies with limited experimental control—together with the replacement of random devices by real-world stories account for much of the difference between the intuitive statistician and Kahneman and Tversky’s heuristics-and-biases program (Gigerenzer and Murray [1987] 2015, Lejarraga and Hertwig 2021, Rizzo and Whitman 2020).

Kahneman’s idea to replace random devices with text problems was hugely successful. By 2020, Peterson and Beach’s (1967) review of 110 studies, which concluded that people are fairly good intuitive statisticians, had been cited 479 times according to Scopus. Up to that date, the article by Tversky and Kahneman (1974) that reviewed just four studies of their own and concluded the opposite had been cited over 15,000 times (Lejarraga and Hertwig 2021). Unlike time-consuming lab experiments, text problems were mobile, inexpensive, and easily scalable. Any lecturer desperate to capture the attention of their students could count on doing so by including one of Kahneman and Tversky’s problems.

HEURISTICS AND BIASES

Edwards concluded that people are conservative Bayesians, that is, they overweight base rates. In contrast, Kahneman and Tversky (1972, 450) reported that people neglect base rates and boldly stated: “man is apparently not a conservative Bayesian: he is not Bayesian at all”. The base rate fallacy was attributed to the representativeness heuristic. Similarly, other biases were attributed to heuristics such as availability and anchoring. The tone was part of the new message. In their first joint publication, on the law of small numbers, Tversky and Kahneman (1971, 107–110) called people’s intuitions a “multitude of sins,” “indefensible,” “self-defeating,” and “ludicrous”. Yet the message was not that people make errors in statistical thinking, which they often do, and that elementary and high schools, medical schools, and law schools should therefore teach everyone how to think in probabilities, which schools rarely do. Their claim was stronger. Cognitive biases were compared to visual illusions, suggesting their inevitability. Influenced by this message, bestsellers such as Dan Ariely’s (2010) *Predictably Irrational*, declared: “we are not only irrational but *predictably irrational*—that our irrationality happens the same way, again and again” (2010, xviii). Kahneman (2011, 29) attributed cognitive biases to a “fictitious” automatic “System 1” that is hardly educable.

The media was influential in spreading the message, as they understood it. Journalists who had shown virtually no interest in the studies of Piaget, Inhelder, Edwards, and others who reported good statistical intuitions trumpeted the cognitive biases story. For instance, a Newsweek feature article asserted that most people are “woefully muddled information processors” and that the list of fallacies of these “suckers” is so lengthy as to “demoralize Solomon” (McCormick, 1987).

But why would people be suckers? Bad decisions had been witnessed since the beginning of humanity and were explained by greed, lust, seduction, group-think, or other desires and anxieties deep in our brain stem. The new message was that the problem is not located in the old parts of our brains but in what makes us *Homo sapiens*, in our very thinking. That message invited a flood of opportunities for blaming the victim. Smoking addiction, compulsive gambling, and the obesity epidemic were attributed to our inner biases, as opposed to the activities of the tobacco, gambling, or fast-food industries. After the financial crisis of 2008, the *Deutsche Bank Research* published an article “Homo Economicus – Or More Like Homer Simpson?” featuring 17 cognitive biases, suggesting these as

causes of the crisis (Schneider 2010). Yet, according to the U.S. Department of Justice (2017), it was the Deutsche Bank that misled investors and contributed directly to an international financial crisis.

In 1994, an Alaskan federal jury awarded \$5.3 billion to fishermen and others whose livelihoods had been devastated by the Exxon Valdez oil spill. When Exxon waged its appeal, it referred to studies that reported jurors' decision processes as "unreliable, erratic and unpredictable" (Hastie 2002, 241) and claimed that "jurors are generally incapable of performing the task the law assigns to them in punitive damage cases" (Zarembko 2003). The results served Exxon well in court. In fact, Exxon had funded the very research, which is published in *Punitive Damages: How Juries Decide* (Sunstein et al. 2002). It became part of a new field known as *behavioral law and economics*. Contributors to the book focused exclusively on the errors of jurors, not on those of Exxon. Eventually, few societal disasters were left that had not been attributed to people's overconfidence, base rate neglect, the conjunction fallacy, or some other mental quirk.

NARROW NORMS

Why were people rational before 1970 and irrational thereafter? In 1991, I posed this question to Ward Edwards in a letter. The occasion was a chapter entitled "Cognitive Illusions" in von Winterfeldt and Edwards (1986), which I thought would address this issue. Surprisingly, however, I found only a brief presentation of Edwards' research on conservatism, followed by a lengthy account of heuristics-and-biases, as if there was no dissent or Edwards' research was simply too old to be true. In response, Edwards sent a long, emotional letter in which he criticized cheap experiments with little control and thoughtless normative standards that portrayed people as irrational: "You are right in taking the illusionists to task for their kneejerk version of the Bayesian position". But why had he not voiced his critique in public? Edwards wrote that he had turned away from the field because "if I had stayed with it, I would have to argue in public with them [Kahneman & Tversky]". In his letter, Edwards referred to Kahneman and Tversky as "my boys" and wrote that one thing he regretted most in his professional life was that he "didn't clamp down on my boys earlier". According to Lewis, Edwards was anxious to avoid any public controversy with them. "Amos was the most terrifying mind most people had ever encountered. They were afraid discussing with him" (Lewis 2017, 155) and "no one wanted to get in a fight with Amos. Not in public!" (Lewis 2017, 319).

NORMATIVE CLAIMS REQUIRE VALID ASSUMPTIONS

Besides the change in method, the key reason why people appeared to be rational before 1970 and not afterwards was Kahneman and Tversky's firm conviction that their text problems have exactly one correct answer and that it is people's fault if they find another. I have designated this the problem of *narrow norms* (Gigerenzer 1996).

Take the Tom W. problem, one of the profession-guessing problems (Kahneman and Tversky 1973). These problems were important to Kahneman, for whom they provided the key demonstration that people neglect base rates and rely on representativeness instead. Participants were given a short description of a person named Tom W., which Kahneman wrote to conform to the stereotype of a computer scientist: "Tom W. is of high intelligence, although lacking in true creativity. He has a need for order and clarity, and for neat and tidy systems..." Then participants were asked how likely it is that Tom W. is a graduate student in computer science, or in the humanities and education, or in other fields. More than 95 percent of them guessed that Tom is more likely in computer science than humanities and education. Arguing that there were more students in the US population in the latter fields, Kahneman and Tversky (1973, 243) diagnosed participants' answers as a case of the base rate fallacy: "the failure to appreciate the relevance of prior probability in the presence of specific evidence is perhaps one of the most significant departures of intuition from the normative theory of prediction".

That is a serious verdict. But before blaming people's intuition for their neglect of base rates, it needs to be made clear whether Tom W. has been randomly drawn from the student population in the US. If, for instance, one meets Tom accidentally at a tech party, it is not a good idea to start thinking about him with respect to the base rates in the US. Yet in the Tom W. problem, no information was provided about how the personality sketch of Tom W. was selected, whether randomly or not (though people could likely sense that the sketch was deliberately constructed). In another profession-guessing problem, the engineer-lawyer problem (Kahneman and Tversky 1973), participants were told that the description was randomly sampled from a pool with 30 engineers and 70 lawyers, but that was not true; once again, the description was made up.

We replicated the experiment but let participants draw the descriptions from a physical urn (Gigerenzer, Hell, and Blank 1988). When they could actually experience random sampling, the base rate neglect largely disappeared and participants paid attention to base rates. The irony is

that participants were more sensitive to random sampling than were the researchers who accused them of committing errors.

Years later, Kahneman (2011) spent an entire chapter on Tom W. and continued to insist that the flaw is in participants' "sins of representativeness" (Kahneman 2011, 151). Here, the reader is given the crucial information: "So far as you know, Tom W was picked at random from the graduate students at the university" (Kahneman 2011, 146). To set the record right, that information was not in the original text problem but added in the book, possibly in response to my critique. I was disappointed to read his account that still did not concede that people were sensitive to base rates if random sampling was actually performed. He was aware of this sensitivity; we had discussed it in both public talks and private conversations.

This omission is not an exception. In *Thinking, Fast and Slow* (2011), Kahneman focused on his own work of the 1970s and 80s, and chose to disregard the considerable body of evidence that casts a shadow of doubt on his rendering of people's cognitive capabilities. Yet progress has been made in understanding people's intuitions since the 1980s. For instance, although the book spends over 60 pages on overconfidence, it makes no mention of the seminal work of Peter Juslin and colleagues, who showed that overconfidence (defined as mean confidence > mean percentage correct) is due to the selection of untypical questions and virtually disappears with random sampling of questions (Juslin, Anders, and Olsson 2000). Other researchers have shown that a second phenomenon also called overconfidence (defined as miscalibration) is largely due to regression to the mean, that is, people's confidence judgments are not systematically biased but have an unsystematic random error (Dawes and Mulford 1996, Erev, Wallsten, and Budescu 1994, Pfeifer 1994; for a summary see Olsson 2014). These follow-up studies reveal the problem of narrow norms, including flaws in researchers' statistical thinking that are misinterpreted as cognitive biases of ordinary people. They also have provided insight into the cognitive processes underlying the motley set of phenomena called overconfidence (Gigerenzer, Hoffrage, and Kleinbölting 1991).

As a final example, consider framing. The work by Craig McKenzie and his group provided evidence that framing is not simply a logical error but an intelligent social inference (e.g., McKenzie and Nelson 2003). Imagine a patient with a serious heart condition who asks a doctor about the risks of surviving surgery. The doctor might answer, "Five years after surgery, 90 percent of patients are alive" (positive frame), or "Five years after surgery, 10% of patients are dead" (negative frame). Because these responses

are logically the same, Kahneman argued that people should not pay attention to the framing. But people—including physicians—do pay attention to it, which Kahneman (2011, 374) took as “embarrassing” and “undeniable” evidence for cognitive biases. Yet studies show that framing conveys useful information. 80 to 94 percent of speakers who have information that alternative treatments would be more dangerous choose a positive frame to make an implicit recommendation, while speakers who believe that alternative treatments are safer choose a negative frame (McKenzie and Nelson 2003). Intelligent listeners then correctly infer what the doctor recommends and act accordingly. The belief that language must be understood in terms of pure logic, without any layers of meaning, and that people who intuitively read between the lines suffer from cognitive biases, is one of the most puzzling narrow norms of the heuristics-and-biases program.

We now know that the framing effect in the Asian disease problem has been misattributed to loss aversion because it disappears when the riskless option is fully specified (Kühberger and Gradi 2011, Mandel 2001). Similar to the sensitivity to random sampling, the fact that people are sensitive to whether one option is fully specified but the other not is hardly compatible with the verdict that attention to framing is purely a bias: “in their stubborn appeal, framing effects resemble perceptual illusions more than computational errors” (Kahneman and Tversky 1984, 343). Much more research has led to progress in understanding the nature of intelligent intuitive inferences, but these two examples should suffice (Gigerenzer 2018).

The general point is: The heuristics-and-biases program takes the discrepancy between judgment and a logical norm as the phenomenon to be explained. Progress was made when researchers abandoned the narrow norms and focused on the judgment itself, not the deviation. The picture that emerges shows fine-grained intuitive inferences that rely on the content rather than on logical structure (see Hahn and Warren 2009, Gigerenzer 2018). Even if this work shows that Kahneman mistook intelligent, context-sensitive behavior for a bias, he could have been proud of having been the inspiration for it. But readers of *Thinking, Fast and Slow* will learn neither about the existence of this research nor about the progress that has been made since the 1970s and 80s.

STATISTICS DOES NOT SPEAK WITH ONE VOICE

Kahneman firmly believed that every text problem he invented has a single correct answer, typically given by probability theory. Yet statistics

does not speak with one voice. To the present day, ever since probability theory emerged in the mid-17th century, mathematicians have debated and disagreed over issues such as whether probabilities are in the mind (epistemic) or in the world (ontic), and whether they involve long-run frequencies (Neyman), propensities (Popper), subjective degrees of evidence (de Finetti), or all three (the classical interpretation of probability). Nor is there agreement on whether Bayes' rule is relevant for science (Fisher: no; Lindley: yes), or whether statistical power is relevant for science (Fisher: no; Neyman: yes) (see Gigerenzer et al. 1989).

One of the most fundamental disagreements concerns whether probability theory applies to singular events. For Neyman and the frequentist school, the answer is "no", for Savage and subjective Bayesians, "yes". In my view, those who want their statements about singular events to follow the laws of mathematical probability are free to do so, but should not force their view upon others. Instead, they need to keep in mind that some of the most influential statisticians would disagree with them. That was not the stance Kahneman and Tversky took. They not only argued that probability applies to singular events, but imposed their view on everyone else.

Consider the Linda problem, which was Tversky's favorite demonstration of cognitive illusions (Lewis 2017, 324).

Linda is 31 years old, single, outspoken and very bright. She majored in philosophy. As a student she was deeply concerned with issues of discrimination and social justice, and also participated in antinuclear demonstrations.

Which of two alternatives is more probable:

Linda is a bank teller

Linda is a bank teller and active in the feminist movement

The majority of undergraduates (85 percent) chose the second alternative. Tversky and Kahneman argued that they commit an error of judgment, the conjunction fallacy. "Like it or not, A cannot be less probable than (A and B), and a belief to the contrary is fallacious" (Tversky and Kahneman 1982, 98). The undergraduates' thinking appeared to indicate a serious defect:

A system of judgments that does not obey the conjunction rule cannot be expected to obey more complicated principles that presuppose this rule, such as Bayesian updating, external calibration,

and the maximization of expected utility. (Tversky and Kahneman 1983, 313)

When participants did not agree that they had made an error, Kahneman invoked the stubbornness of visual illusions, noting that “the sinners seemed to have no shame” (Kahneman 2011, 158).

I was not the first to find fault with placing the blame on people’s thinking, based on the narrow norm that the term *probable* must be understood exclusively as mathematical probability. Commenting on a paper co-authored by Tversky, George A. Barnard, former president of the Royal Statistical Society, remarked: “if we accept it as important that a person’s subjective probability assessments should be made coherent, our reading should concentrate on the works of Freud and perhaps Jung rather than Fisher and Neyman” (Barnard 1979, 171). Even without Freud, the Oxford English Dictionary shows that probable has legitimate non-mathematical meanings, such as “believable,” “acceptable,” and “plausible,” and explicitly marks “relating to or indicating probability” an “obsolete” meaning.

To resolve the question whether the flaw is in the mind or in the norm, one can ask participants about frequencies instead of a single event:

Consider 100 persons who fit the description of Linda. How many of them are:

Bank tellers

Bank tellers and active in the feminist movement

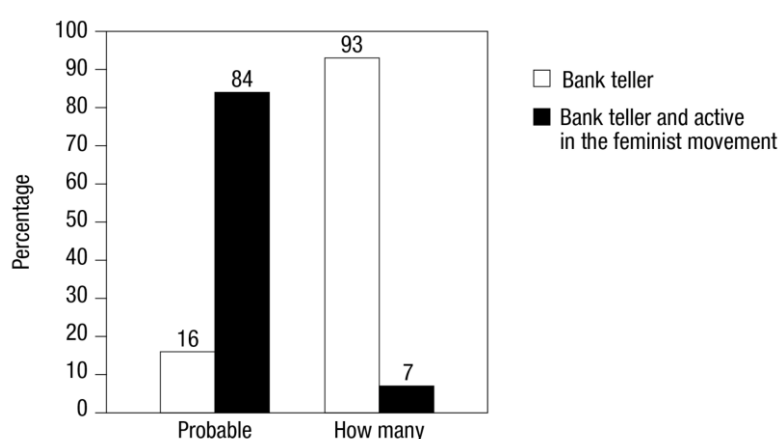


Figure 1: How to make a cognitive illusion disappear. The “conjunction fallacy” (black bars) largely disappears when people are asked about frequencies (how many?) as opposed to the probability of a single event (which is more probable?) (Hertwig and Gigerenzer 1998). For instance, when participants were asked “which is more probable,” as in Tversky and Kahneman’s (1982) study, 84 percent said “bank teller and active in the feminist movement,” replicating the 85% figure in the original study. When participants were asked “how many,” the corresponding figure is only 7%, while the far majority (93%) followed

The “how many” question eliminates the legitimate non-mathematical meanings of the term *probable* so that probability theory applies. Figure 1 summarizes the results, averaged across four studies in Hertwig and Gigerenzer (1998). When participants were asked, “which is more probable?”, the results replicated Tversky and Kahneman’s result perfectly (left side of Figure 1). However, when participants were asked “how many?”, the answers completely changed, and 93% said that there are more bank tellers (right side). Thus, the so-called conjunction fallacy largely disappeared. Fiedler (1988) had earlier found similar results. As a control, we asked participants what they understood by *probable*. The majority understood it to mean plausible, typical, conceivable, or other legitimate non-mathematical terms (Hertwig and Gigerenzer 1998).

The term *and* also has several legitimate meanings in English besides that of a logical conjunction. To understand what *and* means, the human mind once again makes intelligent inferences from content and context. If your neighbor tells you that she has invited friends and colleagues (or bank tellers and active feminists) for dinner, you will likely infer that she has invited members of both groups, not the intersection. Here, people intuitively infer that “and” means logical OR, not logical AND. This inference is not a fallacy. (For the record, Kahneman and Tversky [1996, 587], presented a version of the Linda problem with a frequency question, but changed their original phrasing of the conjunction from “bank tellers and active in the feminist movement” to “bank tellers and active feminists”. In the latter case, as in the example above, people tend to infer that *and* means logical OR. Kahneman and Tversky did not mention that they made this change, but interpreted people’s inference as a fallacy.)

The fascinating ability of the human mind to infer the various meanings of *probable* and *and* from the content and context of a sentence is known as semantic inference. These inferences are necessary to decode the polysemous nature of natural languages, and have been studied by

linguists (e.g., Dulany and Hilton 1991, Grice 1989). The narrow logical norm and the focus on demonstrating biases has prevented researchers from seeing the truly interesting question: How do minds make these intelligent inferences?

In the introduction, I mentioned that Inhelder and Piaget (1964) reported that most children in Geneva understood the conjunction law of probability by age 8. The majority of students at major American universities, in contrast, were reported to commit the conjunction fallacy at an adult age (Tversky and Kahneman 1982, 1983). This apparent contradiction has remained unresolved, if only because Inhelder and Piaget's research is virtually never mentioned. The results shown in Figure 1 (right side) offer an explanation. Inhelder and Piaget showed children various flowers and asked, are there more flowers or more primulas? That is, the children were asked about frequencies, whereas the American students were asked about what is more probable, inviting legitimate non-mathematical interpretations of the question asked.

I had limited success in convincing either Kahneman or Tversky to take a critical look at their norms. At best, they replaced the term conjunction *fallacy* with conjunction *effect*. Or when I addressed the issue in private conversation with Tversky, he responded "let's not talk about the normative issue". Tversky had an excellent training in mathematics, and knew he was on thin ice. In their written response to my critique, however, Kahneman and Tversky (1996, 586) continued to insist that the flaw is in people's mind, and reproached my position as "normative agnosticism".

THE BIAS BIAS

The Tom W. problem, the Linda problem, and various claims of overconfidence and framing bias illustrate a more general phenomenon (Gigerenzer 2018):

Bias bias: The tendency to see systematic biases in behavior even when there is only unsystematic error or no verifiable error at all.

The bias bias also affects studies that use real-world data instead of text problems. Consider the hot hand. Basketball coaches and players frequently recall magical moments where a player is "on fire" or "hot". This temporally elevated performance is known as the "hot hand". After analyzing streaks in free shots, Gilovich, Valone, and Tversky (1985) concluded that this belief is a mere illusion, which they named the *hot hand*

fallacy. Coaches and players who cast doubt on this diagnosis were subsequently accused of being “unwilling even to consider the possibility that their strongly held beliefs might be wrong” (Thaler and Sunstein 2008, 31). That seemed to provide more evidence that cognitive illusions resemble stubborn visual illusions. “The hot hand is a massive and widespread cognitive illusion,” Kahneman (2011, 117) confidently asserted. However, Miller and Sanjuro (2018) showed that the flaw lies not in the minds of coaches and players but in the statistical reasoning of the researchers. Ironically, this flaw is similar to the law of small numbers, the topic of Tversky and Kahneman’s (1971) first joint publication. The researchers had mistakenly assumed that the properties of streaks (which are small samples) are the same as those of the population. A reanalysis of the free shot data showed that the hot hand indeed exists (Miller and Sanjuro 2018). Alongside people’s attention to random sampling and to the polysemy of the word *probable*, belief in the hot hand is another case where people’s fine-tuned intuitions are mistaken for fallacies. In this case, even Tversky himself was fooled by probability.

A similar argument has been made for people’s alleged misconceptions of chance. Yet the properties of small samples are again not representative of the population. Take, for instance, the question of which sequence (H = head, T = tail) one is more likely to encounter when flipping a fair coin:

HHHH or HHHT?

Most people say that the sequences HHHT is more frequent, while researchers have asserted that both are equally likely and that people misconceive chance. But at closer look, if someone flips a fair coin 20 times, the probability that they will see at least one HHHH is about 50%, whereas the chance of a HHHT is around 75% (Hahn and Warren 2009). The probability of encountering an HHHH is equal to that of an HHHT only if $k = n$, where k is the length of the sequence and n the total number of times a coin is flipped. Once again, claims about statistical intuitions require a close examination of the situation (here, whether $k < n$ or $k = n$). This analysis of the conditions in which an inference, or a heuristic, is successful or not is the topic of the study of ecological rationality (see below).

My general point is not that statistical thinking can be difficult (although it often is). Rather, it is the overconfidence of researchers that they know the right answer and all others are wrong. More humility, curiosity, and willingness to give people the benefit of doubt would be helpful for making progress.

BERNOULLI'S ALLEGED ERROR

Kahneman generally wrote in a modest and self-reflective way, pointing out that he himself committed the statistical errors they investigated. Nevertheless, he was fully convinced that he knew the correct answers. In all likelihood, his least modest claim was to accuse the Swiss mathematician Daniel Bernoulli of having committed an error (Kahneman 2002b, 2011). Allow me to defend Bernoulli here. After all, this accusation has been repeated and, to the best of my knowledge, gone unanswered (apart from a technical report by Charles-Cadagon 2015). Here is what Kahneman wrote about Bernoulli:

The language of Bernoulli's essay is prescriptive – it speaks of what is sensible or reasonable to do – but the theory is also intended to describe the choices of reasonable men (Gigerenzer et al., 1989). As in most modern treatments of decision making, there is no acknowledgment of any tension between prescription and description in Bernoulli's essay.

The idea that decision makers evaluate outcomes by the utility of final asset positions has been retained in economic analyses for almost 300 years. This is rather remarkable, because the idea is easily shown to be wrong; I call it Bernoulli's error.

Bernoulli's model is flawed because it is *reference-independent*: it assumes that the value that is assigned to a given state of wealth does not vary with the decision makers' initial state of wealth (Kahneman 2002b, 460).

When I read this claim, I was puzzled. Kahneman had asked me in an email about Bernoulli's understanding of utility, and I responded that Bernoulli ([1738] 1954) did not distinguish between prescription and description, which was common for the Enlightenment mathematicians. I referred him to our book *The Empire of Chance* (Gigerenzer et al. 1989), which he cited. How did he come to believe that Bernoulli's utilities do not vary with decision makers' initial state of wealth? The initial wealth of the decision maker can namely be found in Bernoulli's equations. His utilities are relative, that is, reference-dependent, and not attached to the final state of wealth, as Kahneman asserted.

In fact, Bernoulli defined utilities on gains and losses. For instance, referring to his figure where B is the initial state of wealth, and po and PO are a loss and a gain of equal amount, respectively, Bernoulli wrote: "it is

clear that the disutility *po* which results from a loss will always exceed the expected gain in utility, *PO*" ([1738] 1954, 29). When introducing prospect theory, Kahneman and Tversky (1979, 279) proposed the same effect, "losses loom larger than gains," but incorrectly claimed that "Markowitz (1952, 276) was the first to propose that utility be defined on gains and losses rather than on final asset positions". Two of prospect theory's major claims—that utility is reference-dependent and that losses loom larger than gains—are in fact in Bernoulli's theory.

Just to be sure, I asked Gérard Jorland, a historian of science and expert on Bernoulli's utility theory (e.g., Jorland, 1987) for his opinion. Jorland responded: "True, Kahneman is wrong when he says that Bernoulli 'assumes that the value that is assigned to a given state of wealth does not vary with the decision makers' initial state of wealth'" (email, September 1, 2005). I am still perplexed why Kahneman made these incorrect claims. Perhaps he thought that the version of utility theory presented in Kahneman and Tversky (1979, 263)—which does not include initial wealth—was the same as Bernoulli's theory. But what I find it even harder to understand is the next step, to wrongly criticize Bernoulli that his utilities were reference-independent. Even more puzzling is why these claims were so rarely disputed.

Similarly, Rakow (2010) pointed out that Edwards (1962) and, before him, Knight (1921) had already identified changes in wealth as the basis of utility, as in prospect theory, in addition to non-linear decision weights. Kahneman (2010, 463) reacted oddly and defensively: "This is not how science is done. Science is essentially a conversation in which people respond to what others have most recently said". He continued: "Reading Knight would not have helped us at all—we would not have recognized that what he said meant the same thing as what we said, and indeed it did not (because what we said in prospect theory was a reaction to the idea that utility is attached to final states, which was dominant in decision theory at that time, but had of course been asserted by Bernoulli in 1738)" (Kahneman 2010, 463–464). The fiction of Bernoulli's error became the protective fortress around Kahneman's faith that prospect theory was more novel than it is.

VAGUE HEURISTICS

One of Tversky's important contributions is elimination-by-aspect, a mathematical model of a heuristic for choice (Tversky 1972). It proposes a sequential process in which options are eliminated, aspect by aspect. Interestingly, such a "lexicographic" process is incompatible with a utility function. When Tversky joined Kahneman, he stopped working on *models*

of heuristics. Instead, the two of them introduced three *labels*—availability, representativeness, and anchoring—to explain deviations from logical norms. With such labels, it is impossible to know whether these can be mapped into a utility function, or what other interesting properties they have.

It is understandable that when these terms were first proposed in the early 1970s, they were only loosely characterized. Yet over two decades later, when Kahneman and Tversky (1996) replied to my critique, these one-word explanations were still undefined with respect to the cognitive processes that underlie them and the antecedent conditions that elicit or suppress them. I was not the first to point out that this vagueness allows virtually every behavior to be explained post hoc (Einhorn and Hogarth 1981, Shanteau 1989).

Indeed, representativeness has been used to explain both a phenomenon and its antithesis, that is, account for both A and non-A (Ayton and Fisher 2004). Consider the gambler's fallacy and the hot hand fallacy. When in a random sequence a run is observed (such as a series of reds on the roulette wheel), people tend to believe that the *opposite* result (black) will come up next. Tversky and Kahneman (1974, 1125) wrote that this intuition—the gambler's fallacy—is due to representativeness. Now consider the opposite phenomenon. When in a random sequence a run is observed (e.g., a series of hits scored by a basketball player), people tend to believe that the *same* result will come up next. Gilovich, Vallone, and Tversky (1985, 295) explained this intuition—which they called the hot hand fallacy—by representativeness as well.

Similarly, one-and-the-same phenomenon has been explained post hoc by several different labels. Take, for instance, Kahneman and Tversky's manuscript "On the Reality of Cognitive Illusions: A Reply to Gigerenzer" (the subtitle was deleted in the final version), for which I was a reviewer. In the first paragraph of the original manuscript, they illustrated the availability heuristic, noting that "a judgment of the prevalence of suicide in a community is likely to be mediated by the number of instances that come to mind". In my review, I pointed out that availability had previously been said to be mediated by the ease with which instances come to mind. Which is it, number of instances or ease? In reaction, the published version reads: "is likely to be mediated by the ease with which instances come to mind" (Kahneman and Tversky 1996, 582).

The problem is not that a statement was corrected; that is part of the review process for all authors. What poses a problem are the multiple and exchangeable meanings of the term *availability*, which have not been resolved.

Are ease and number of instances the same? The classical study on availability is the letter-R experiment (Tversky and Kahneman 1973), prominently described at the beginning of *Thinking, Fast and Slow* (Kahneman 2011). For Kahneman, this study was the key demonstration of the availability heuristic. Participants were asked whether there are more words with the letter R in the first or in the third place. Most chose the former, which was explained by availability: Words with the letter R in the first place are more readily available. However, there was no independent measure of availability; the explanation was post hoc. We replicated the letter-R experiment but additionally measured for each individual the time (ease) it took for the first instance to come to mind and also the number of instances that came to mind (Sedlmeier, Hertwig, and Gigerenzer 1998). Then we tested whether individuals' time and number of recalled instances actually predicted their word frequency estimates. Neither ease nor number did so, nor was there any substantial correlation between these two measures of availability. This example illustrates the peril of plausible post-hoc stories and the need for measuring the postulated heuristic processes independently.

Ease and number are not the only meanings of availability. In behavioral law and economics, for instance, availability means many different things: ease, imagined ease, number, recency, salience, and vividness (Gigerenzer 2006). What is needed are process models of heuristics, as in Tversky's earlier work, that clarify how the heuristics operate, rather than merely offering post hoc explanations for a wide range of phenomena.

However, such progress did not materialize. In a postscript to their 1996 article, Kahneman and Tversky continued to defend their "approach for not fully specifying the conditions under which different heuristics control judgment" (Kahneman and Tversky 1996, 591). Gestalt psychology, they argued, did not specify its processes either, and to legislate process models would be "unwise". This was an un-Tverskyian response. Tversky had favored precise mathematical models in his seminal earlier work on similarity and measurement theory. According to Lewis (2017, 329), Tversky had actually tried to define "representativeness", but Kahneman "had wanted to keep the definition vague".

PERSONAL RECOLLECTION

My first personal contact with Kahneman was in February 1990. At that time, I was a fellow of the Center for Advanced Studies in Stanford, and had just given a talk at the Department of Psychology at Stanford University, the department where Tversky had been working for decades. The

title of my talk was “Beyond heuristics and biases: How to make cognitive illusions disappear” (see Gigerenzer 1991, 2024). Two weeks later, Kahneman contacted me: “I heard you had a very successful colloquium at Stanford. Would you be willing to give one or two talks at Berkeley?”

Kahneman was trying to start a conversation. That was very different from how defensively and angrily Tversky had reacted to my talk.

I went to Berkeley in March. Before my talk at the Department of Psychology, Kahneman and I took a walk together. I was taken by his charming personality and his openness to discussing ideas. He requested 10 minutes to respond after my talk, after which I had 5 minutes to reply. I grew fond of this format. A public exchange of ideas offers the audience a rare opportunity to hear both sides at the same time and then move forward from there.

On the way to the lecture hall, Kahneman introduced me to the department secretary, adding: “This is the German psychologist who will destroy me in an hour”. I mention this not only because the remark puzzled me but because I heard him use war metaphors ever more frequently.

Yet Kahneman kept the public conversation going. He invited me again to give a talk using the same format for the Cognitive Science Group at Berkeley. After he had finished his 10-minute reply, he said something remarkable in public: “Gerd is much younger than Amos and me, so he will win”. Once again, I was puzzled about this “Wild West” metaphor of scientific debate. When we were walking to a restaurant in Berkeley for dinner, Kahneman turned to me and said: “When we were young, we fought Ward Edwards and we won, and we misrepresented him unfairly. Now we are old and this is happening to us. I knew that someday someone would come and fight us, and now someone has come”. As you may infer, the implication was that I misrepresented him unfairly. Yet I was not out to fight people, but to challenge ideas and improve them—what I see as the aim of science.

The intellectual exchange between Kahneman & Tversky and myself has been dubbed the “rationality wars” (Samuels, Stich, and Bishop 2002). I am not partial to the term *war*, given that I tried hard to separate scientific disagreements and personal respect (Gigerenzer, 2024). It is easy to contest someone’s ideas if you dislike the person, but it is emotionally demanding to disagree with mutual respect. It was not always easy for either side, but Kahneman and I both did our best.

To illustrate, let me quote from a letter Kahneman sent me on December 28, 1992. The occasion was a keynote I gave at the Judgment and Decision-Making Conference in St. Louis in November 1992, where he again had a 10-minute response, followed by my reply. After our joint

event, I had sent him a letter to thank him for his continuing willingness to openly engage in a debate over the issues on which we disagreed.

Dear Gerd,

Thank you for your gracious letter ...

Let me begin by a personal remark. I was impressed at the St. Louis meeting by how popular you are among your Chicago colleagues, including some old friends and associates of mine. Indeed, I too have found you consistently charming as well as interesting in our interactions, although I was not always pleasant myself. What then is wrong? As I reflected on it, the titles of your papers and talks have a lot to do with the atmosphere of sharp controversy that surrounds them, and with the unusually aggressive stance that I have found myself taking in responding to you. Announcing the death of someone else's paradigm is permissible in the American context, but far less common than in Europe—it may accordingly be taken as more pointedly aggressive than you seem to intend it. Of course, announcing the death of someone else's paradigm with poise, charm, and authority is even more uncommon. I observe with some amusement that you seem to benefit, in the American context, from a subtle cultural misunderstanding which also benefits Israelis—most certainly including both Amos and myself: the confident assertion of extreme opinions is both valued and inhibited in the American academic culture, and the rewards for taking the risk are rich if you get away with it. You are obviously getting away with it, as we have also, although I am quite sure that you are mostly wrong and we are mostly right! As you can see, your popularity among my friends and the graciousness of your letter have had quite an impact on my personal attitude to you, and to the controversy. [...]

D. Kahneman

I quote from this letter at length to give a sense of our joint effort to maintain personal respect during an intense scientific debate. It was worth the emotional difficulties. We practiced this format of public exchange—talk, response, reply—three times in the early 1990s. The initiative came from Kahneman, which speaks for his intellectual openness. His idea of joint talks could serve as a model for how to handle scientific controversies constructively.

Kahneman also introduced a written version of this approach, which he called “adversarial collaboration,” where he coauthored papers with

scholars whom he fundamentally disagreed with. One such collaboration was with my former student Ralph Hertwig (Mellers, Hertwig, and Kahneman 2001), another was with Gary Klein, who studied expert intuition and disagreed with Kahneman's debunking of it (Kahneman and Klein 2009). In Kahneman's own words, "more efficient procedures for the conduct of controversies will be part of my legacy" (2002a).

While Kahneman made a deliberate and emotionally demanding effort to engage with his critics, Tversky's style was more confrontational. I experienced this firsthand, both in the heated debate at the reception following my talk at Stanford and in the letters Tversky sent me. As Lewis (2017, 337) observed, "Danny was always the appeaser, Amos the bully". The day after my talk, Tversky and I met in his office to discuss our differences for an hour, which turned into two, without making any progress. He pushed the fact that people make errors but refused to discuss the normative standards. At one point, he asked me, "Why don't you join Danny and me?" I appreciated the gesture, but I was committed to pursuing a deeper understanding of the usefulness of heuristics and the nature of rationality. I did not want to join a team that, in my view, continued to insist on vaguely defined heuristics and overly narrow normative standards.

According to Kahneman (2002a), he and Tversky had agreed in the early 1980s not to respond to critics. However, around 1994, Tversky insisted that Kahneman join him in replying to my arguments. Kahneman agreed, even though by then, as he later admitted, their friendship had ended. The resulting article, "On the Reality of Cognitive Illusions", was published in *Psychological Review* in 1996. My reply appeared in the same issue, titled "On Narrow Norms and Vague Heuristics" (Gigerenzer 1996). The exchange included postscripts, added on Tversky's insistence. As Lewis noted, "Amos didn't merely want to counter Gigerenzer; he wanted to destroy him" (2017, 335). After reading that, I felt my efforts to engage Tversky in a productive dialogue had ultimately failed. I can only speculate that Tversky was not used to anyone challenging his ideas on his home turf in the presence of the assembled Stanford faculty, and never forgave me for it.

Looking back, I deeply appreciate Kahneman's willingness to endure the emotional strain of engaging in open disagreement with colleagues while striving to maintain personal respect.

KAHNEMAN'S SOLUTION TO THE CONFLICT: TWO SYSTEMS

Eventually, Kahneman proposed a solution to our controversy over cognitive illusions. The solution was not to reconsider the narrow normative

standards which made people appear lacking in rationality. Instead, he continued to insist that the text problems had a single correct answer and that the problem was to be found in other people's minds. Around the turn of the millennium, Kahneman adopted a dual-systems framework, proposed earlier by Sloman (1996) and other psychologists.

The dual-process model, for Kahneman, offered a way to reconcile the allegedly stable cognitive biases with the fact that those same biases could largely be made disappear (e.g., Figure 1). In his own words:

A young colleague and I recently reviewed the experimental literature, and concluded that the empirical controversy about the reality of cognitive illusions dissolves when viewed in the perspective of a dual-process model (Kahneman & Frederick, 2002). The essence of such a model is that judgments can be produced in two ways (and in various mixtures of the two): a rapid, associative, automatic, and effortless intuitive process (sometimes called System 1), and a slower, rule-governed, deliberate and effortful process (System 2) (Sloman, 1996; Stanovich and West, 1999). System 2 'knows' some of the rules that intuitive reasoning is prone to violate, and sometimes intervenes to correct or replace erroneous intuitive judgments. Thus, errors of intuition occur when two conditions are satisfied: System 1 generates the error and System 2 fails to correct. In this view, the experiments in which cognitive illusions were "made to disappear" did so by facilitating the corrective operations of System 2. (Kahneman 2002a)

I found Kahneman's chosen solution surprisingly weak. If people erred, blame was put on the intuitive System 1 and on the rational System 2's failure to intervene. If people got the answer right, it was because System 2 had corrected System 1. This framework made it possible to explain every conflicting result after the fact.

I cannot imagine that Tversky would have agreed to resolve the debate over the nature of cognitive illusions by appealing to two fictional systems, one that makes the error and the other not. As Allen Newell had warned, with explanations based on binary oppositions such as nature/nurture, clarity is never achieved; instead "matters simply become muddier and muddier as we go down through time (Newell 1973, 288–289). Along with Terry Regier, I argued that the dichotomies of which Slo-

man's dual-system account consisted were too vague and general, offering a tempting illusion of explanatory power while neglecting to actually model any cognitive processes (Gigerenzer and Regier 1996).

When I studied, in a different context, the 19th-century idea of female versus male intelligence, I was surprised to find a similar list of dichotomies. Women were said to be intuitive, associative, fast in thinking, and often wrong, just like System 1. In contrast, men's thinking was understood as deliberate, slow, rule-governed, rational, and mostly right. Moreover, men were expected to correct their wives' actions, just as System 2 is supposed to watch over System 1 (Gigerenzer, 2023). An antiquated list of male-female dichotomies appears to have made a stellar revival in a new domain, now applying to each individual regardless of gender.

The two-systems account allowed researchers to reconcile contradictory findings without having to re-examine the narrow normative standards that underpinned the original claims of cognitive illusions. Even Kahneman himself was uneasy with his proposal. He acknowledged that there is no neurological evidence for such two systems. Pre-empting criticism, he wrote: "the two systems do not really exist in the brain or anywhere else" (Kahneman 2011, 415). Yet he made them the centerpiece of *Thinking, Fast and Slow*, using System 1 and System 2 to explain virtually every cognitive bias, with System 1 as the unreliable, intuitive process and System 2 as the lazy overseer.

What troubles me most is how readily many psychologists embraced this narrative. Instead of focusing on modeling actual mental processes, part of the field, in my view, descended into a low point in theorizing. Even respected economists have followed suit. Shleifer (2012, 6), for instance, wrote that "System 2 is asleep" and appeared to treat this as a scientific explanation.

HOW TO MAKE PROGRESS

In his biographical essay, Kahneman (2002a) reflected on what he learned from the controversy about heuristics and biases: "My long-standing failure to associate specific benefits to the concept of representativeness was a revealing mistake". More generally, for all the care taken in highlighting errors, Kahneman and Tversky never reported a single case in which a heuristic demonstrably worked well. Part of his legacy is to answer the question, how can we move forward toward a systematic understanding of the benefits of heuristics?

Progress requires a step Kahneman did not want to take: developing algorithmic models of heuristics. These models make the structure of

heuristics explicit, allow for testable predictions, and move the field beyond post-hoc explanations.

ECOLOGICAL RATIONALITY

In their first papers, Kahneman and Tversky already acknowledged that heuristics can often be useful though *sometimes* lead to severe and systematic errors. But they never worked out what “sometimes” meant. The study of ecological rationality addresses this crucial question: Under what conditions can a heuristic—or any decision strategy—be expected to perform successfully? (Gigerenzer, Hertwig, and Pachur 2011).

Answering that question requires algorithmic models and clear goals such as predictive accuracy. Ecological rationality builds on Herbert Simon’s (1990) famous scissors analogy: Understanding behavior requires studying both the cognitive and the environmental “blade”. Only by examining how these two blades interact can we understand why a strategy works or fails. The study of ecological rationality fleshes out Simon’s analogy and addresses Kahneman’s afterthought about understanding the benefits of heuristics.

Take, for example, availability in the form of *recency*. This heuristic can be formalized as a simple algorithm: To make a prediction, use only the most recent data point, and ignore all prior information (Katsikopoulos et al. 2022). In the heuristics-and-biases tradition, ignoring data—such as base rates—has been unconditionally interpreted as a cognitive flaw. While this may hold in stable, well-defined settings (what Savage [1954/1972], called “small worlds”), in uncertain, unstable environments (“large worlds”), simple heuristics can outperform complex models.

Consider the problem of predicting demand. A research team around economist Joseph Stiglitz built an agent-based model to simulate complex, evolving macroeconomic systems. In this volatile environment, the recency heuristic predicted demand better than sophisticated macroeconomic forecasting models. The authors concluded: “fast and frugal robust heuristics may not be a second-best option but rather ‘rational’ responses in complex and changing macroeconomic environments” (Dosi et al. 2020).

Another example is predicting seasonal flu. Between 2008 and 2015, Google Flu Trends used big data algorithms to forecast flu-related doctor visits. But viruses create an inherently unstable situation; events like the swine flu can upend past patterns. When we tested the recency heuristic,

it consistently outperformed Google's algorithm, including all of its updates, for every year of that period (Katsikopoulos et al. 2022).

Importantly, these examples are drawn from real environments characterized by uncertainty—not from textbook problems with a “correct” logical answer. More broadly, research in ecological rationality has identified a set of environmental structures that heuristics can exploit to make more efficient and accurate predictions than complex models (see Gigerenzer 2021, Katsikopoulos 2023).

The broader lesson is this: The rationality of heuristics is not logical but ecological; understanding why heuristics work requires knowledge of how cognition and environment interact. Heuristics enable good decisions in situations of uncertainty, that is, where the state space of possible future events and their consequences are not known or knowable. In contrast, the heuristics-and-biases program continued to seek explanations within the mind—via internal consistency, overconfidence, loss aversion, or System 1 versus System 2. But only by analyzing the structure of the external world can it be determined when a heuristic, overconfidence, or loss aversion will be successful or not.

No strategy is optimal in all situations. The challenge—and the opportunity—is to identify when and why heuristics succeed.

LESS CAN BE MORE: THE BIAS-VARIANCE TRADE-OFF

Kahneman's understanding of the benefits of heuristics was consistent with the *accuracy-effort trade-off* (Payne, Bettman, and Johnson 1993): The less effort one takes, the less accurate one will be. Accordingly, heuristics, because they save effort, pay a price in terms of accuracy. This trade-off, however, provides only a limited understanding of the benefit of heuristics, reserving them for unimportant decisions where the benefit of saving time can outweigh accuracy. As far as I can see, that has been the dominant understanding of the benefit of heuristics in the heuristics-and-biases program.

It is true that heuristics generally save effort, but not that this necessarily leads to less accurate judgments than do more effortful rules. The bias-variance trade-off, known from machine learning (Geman, Bienenstock, and Doursat 1992), helps to understand these *less-is-more effects* in a systematic way. It clarifies that the source of error is not only bias (the difference between the true value and the mean estimate generated by a model) but also variance (the variability of a model's estimates

around its mean estimate). Heuristics such as the recency heuristic, recognition heuristic, or $1/N$ (equal allocation) use zero free parameters, which may increase bias but reduces error from variance to zero (Gigerenzer and Brighton 2009). Thus, the goal is not to reduce bias to nil, but to find a good trade-off between bias and variance.

By the late 1990s, my research group had demonstrated cases where fast-and-frugal heuristics predicted more accurately than more complex models such as multiple regression (Gigerenzer, Todd, and the ABC Research Group 1999). These cases are not reconcilable with the assumption of a general accuracy-effort trade-off. Rather, they demonstrate the existence of less-is-more effects, that is, situations where less effort leads to more accurate predictions. The relationship between accuracy and effort is typically inversely U-shaped, not monotonically increasing. That is, increasing effort or model complexity improves accuracy only up to a point, after which further increases will reduce accuracy. This inversely U-shaped curve is explained by the bias-variance trade-off. It provides a rational explanation for the benefit of heuristics, one that goes beyond the accuracy-effort trade-off.

KAHNEMAN'S LEGACY

That said, without Kahneman's work, we would not be where we are. The research on heuristics and biases has had an impact on the social sciences that was rarely matched by any other program of its time. It inspired countless scholars to build on the foundation that Kahneman and Tversky laid and to explore paths they left open.

Opportunities Taken

The heuristics-and-biases program made seminal contributions in several key respects:

- It brought psychology into economics (once again).
- It brought heuristics to the attention of social scientists.
- It brought statistical thinking to the attention of the general public.

These are lasting achievements. The rise of behavioral economics stands as one illustration. My own research on fast-and-frugal heuristics was inspired by Kahneman and Tversky's groundbreaking work.

Opportunities Not Taken

At the same time, the program also missed opportunities, such as:

- Bringing psychology into economics not only as a source of irrationality but also as a source of genuine intelligence.
- Moving beyond labels and dichotomies toward formal models of heuristics.
- Shifting from the content-blind ideal of logical rationality to the study of ecological rationality.
- Moving from seeking errors in statistical thinking to actively teaching statistical thinking.

In short, the program largely remained stuck in the framework of the 1970s. Progress can be made by seizing these missed opportunities. For example, the citation for Kahneman's 2002 Nobel Memorial Prize in Economic Sciences credits him "for having integrated insights from psychological research into economic science, especially concerning human judgment and decision-making under uncertainty". Yet the role psychology was allowed to play in his work was largely to explain why people deviate from rational norms. Kahneman and Tversky remained, in many ways, conservative defenders of economists' notions of logical rationality—taking these more seriously, perhaps, than many economists did themselves.

The widespread claim that cognitive illusions, like visual illusions, are uneducable has slowed down progress. Kahneman continued to claim that "it takes an enormous amount of practice to change your intuition" (cited in Bond, 2009, 1191). Research, in contrast, has shown that we can successfully and quickly teach statistical reasoning to children and adults. As studies in China (Zhu and Gigerenzer 2006) and Germany (Gigerenzer et al. 2021) have demonstrated, by using intuitive representations of numerosity, the majority of ten-year olds can already reason the Bayesian way. And a two-hour course teaching intuitive techniques such as "natural frequencies" can boost this competence (Jenny, Keller, and Gigerenzer 2018) and sustain it for a long time (Sedlmeier and Gigerenzer 2001). I have taught over 1,000 gynecologists in their continuing medical education, as well as law students and US federal judges. At the beginning of the courses, most did not understand how to evaluate the chance that a woman with a positive screening mammogram actually

has breast cancer, or the chance that a defendant whose DNA matches with a trace found on the victim is actually the source of the trace. Yet they quickly learned to think statistically (Gigerenzer 2014).

The general problem is that educational systems continue to devote scant attention to teaching statistical thinking, and if it is taught, mostly in a way that is not intuitive. In most parts of the world, children are taught the mathematics of certainty, not uncertainty. Even most medical schools do not take sufficient efforts to teach future generations of doctors how to understand health statistics (Gigerenzer et al. 2007). Unlike visual illusions, errors in statistical thinking are avoidable. Only a few hours of practice can make all the difference.

CIVILITY AND ADVERSARIAL COLLABORATION

Let me end with what may be Kahneman's most important legacy: his willingness to engage in what he called "adversarial collaboration". One can hardly overestimate the emotional strain it caused him. His openness to debate began with the three joint talks we had in the early 1990s and continued through the adversarial collaborations he initiated with several of his critics.

Learning to separate the personal from the intellectual—to debate an issue without assuming malicious intentions on the other side—is one of the most virtuous and difficult achievements in science. The history of science is full of stories of those who failed to do so. Renaissance mathematicians once dueled over solutions to cubic equations, and Newton famously broke Leibnitz's heart during their dispute over who invented calculus. That rivals eventually learned to speak to each other with respect, and even to cooperate, is a relatively recent development in the sciences (Daston 2023).

Economist Paul Romer once remarked that the trouble with macroeconomics is that economists are more committed to their friends than to facts. This tendency to please colleagues, he argued, contributed to over three decades of intellectual regress (Romer 2016). Yet this dilemma is not confined to macroeconomics. Being open to critical arguments and ready to change one's position remains a rare virtue. Kahneman himself often lamented that people do not change their minds. And yet I am grateful that he tried, and in doing so, set an example of how to engage in debate with civility and integrity.

My last contact with Kahneman was by email, in April 2023, while I was visiting my daughter's family in Princeton and hoped to see him. He replied that he had already moved to New York and invited me to visit

him there. My regret is that I was too busy savoring the little time I had with my grandchildren to take him up on the invitation, and never saw him again.

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